

Project Initialization and Planning Phase

Date	21-09-2026
Team ID	MCA
Project Title	Strategic Employee Attrition Intelligence Dashboard
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Quantify overall and segment-level attrition rates (department, job role, work mode); identify the strongest predictors of attrition (burnout, overtime, compensation growth, tenure, performance); build an interactive Power BI dashboard for HR and leadership decision-making; provide AI-assisted insights for root-cause analysis; and deliver actionable recommendations to reduce attrition and improve workforce wellbeing.
Scope	Covers analysis of all 5,000 employee records — demographics, compensation, tenure, workload, engagement, performance, and attrition outcomes — including segment-level breakdowns by department, job role, and work mode, and delivery of an interactive Power BI dashboard and report.
Problem Statement	
Description	895 of the organization's 5,000 employees (17.9%) have left, and HR cannot currently pinpoint which factors — compensation, workload, engagement, career progression, or work arrangement — most strongly predict attrition.
Impact	Without this understanding, HR cannot prioritize retention spend or design targeted interventions, which drives up recruitment and training costs, increases workload on remaining employees, and reduces overall productivity.

Proposed Solution	
Approach	Analyze employee demographics, compensation, tenure, workload, engagement, and performance data to quantify attrition rates and identify the strongest predictors; build an interactive Power BI dashboard and report; layer in AI-assisted insights to support root-cause analysis; and translate findings into evidence-based retention recommendations.
Key Features	Interactive Power BI dashboards segmented by department, job role, and work mode; predictor analysis covering burnout, overtime, compensation growth, tenure, and performance; AI-assisted root-cause insights; and actionable recommendations to reduce attrition and improve workforce wellbeing.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Operating System	OS compatibility	Windows 10 or Windows 11 (64-bit)
Computing Resources	Processor specification	Intel Core i3 / AMD Ryzen 3 or higher
Memory	RAM specification	Minimum 8 GB
Storage	Free disk space required	At least 5 GB
Display	Screen resolution	1366 × 768 resolution or higher
Software		
Database	Database engine	MySQL / SQL Server or any compatible SQL database
SQL Tool	SQL client / management tool	MySQL Workbench / SQL Server Management Studio
Power BI	Business intelligence tool	Power BI Desktop (latest version recommended)

Data		
Data	Employee HR dataset — demographics, compensation, tenure, workload, engagement, performance, and attrition outcomes (29 attributes per employee, incl. department, job_role, monthly_income, salary_hike_pct, years_at_company, overtime_hours_per_week, burnout_score, engagement_score, work_life_balance_score, performance_rating, uses_ai_tools_at_work, attrition)	5,000 employee records; 895 attrition cases (17.9%); loaded into Power BI via SQL/CSV source